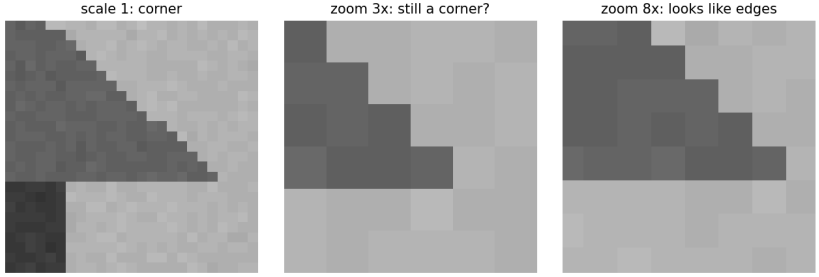
Industrial Vision

Lecture 6 - Blob Detection & Scale Space

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Where we are

The Harris detector provided corners that are invariant to translation, to rotation and to an additive change $I+a$, with a change of gain merely rescaling the response. It failed, however, on **one** transformation.



If a corner is magnified, then within the **fixed** Harris window it degenerates into straight **edges**. Whether a structure is a corner therefore depends on the scale at which it is examined.

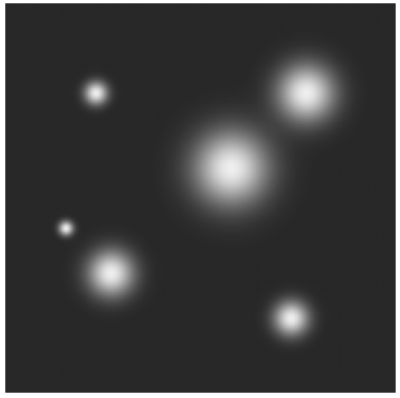
The cause of the difficulty is the fixed window. The remedy is to allow the window to vary in size, searching over a scale axis and letting each structure indicate the scale at which it is most pronounced.

Today we develop this machinery on a feature simpler than a corner, namely a **blob**. In the next lecture, on SIFT, it is extended into a complete detector and descriptor.

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Blobs occur at every scale

A **blob** is a compact region that stands out from its surroundings, being either brighter or darker than the region enclosing it.



Blobs of many different radii coexist within a single image.

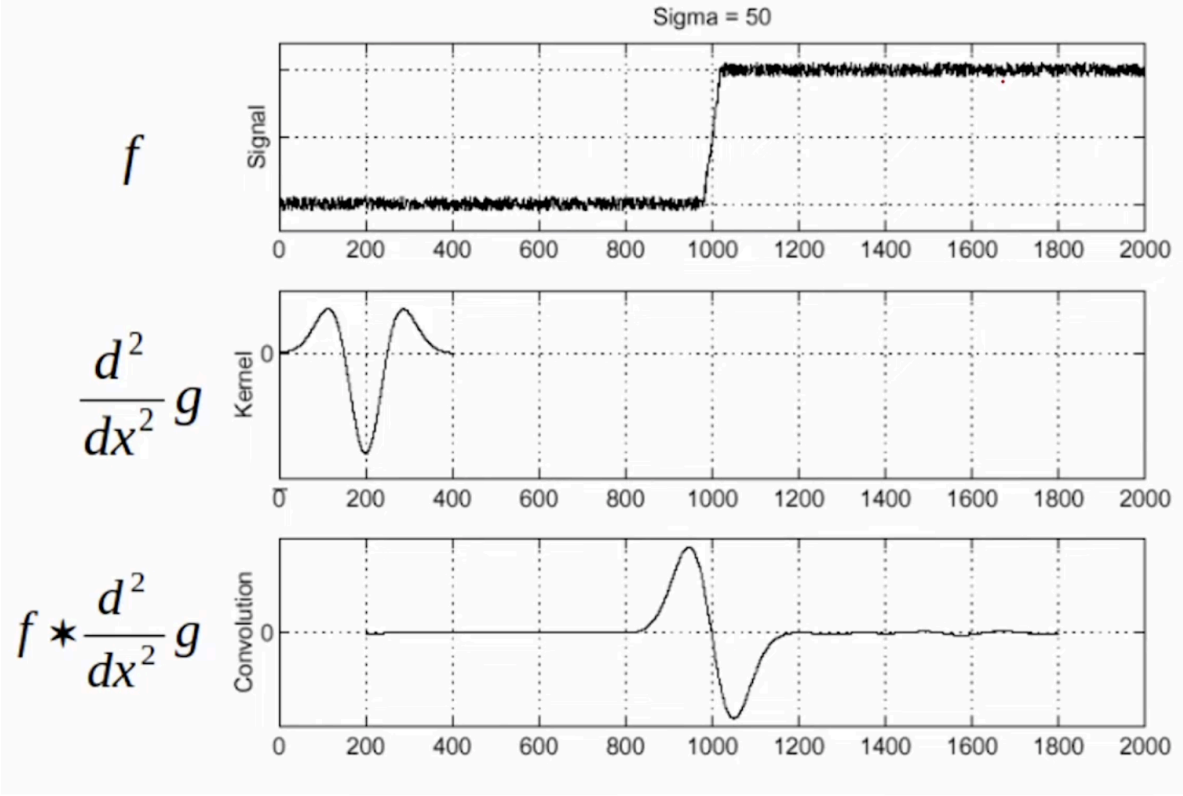
Consider a **probe** applied to the image.

- A **small** probe resolves small blobs, but a large blob produces almost no response.
- A **large** probe responds strongly to a large blob, but it can no longer resolve two small blobs lying close together.

The size of the probe must therefore **adapt**. For each structure we find the scale at which the probe fits best, and that scale is itself a property of the structure and is worth recording.

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LoG on step function

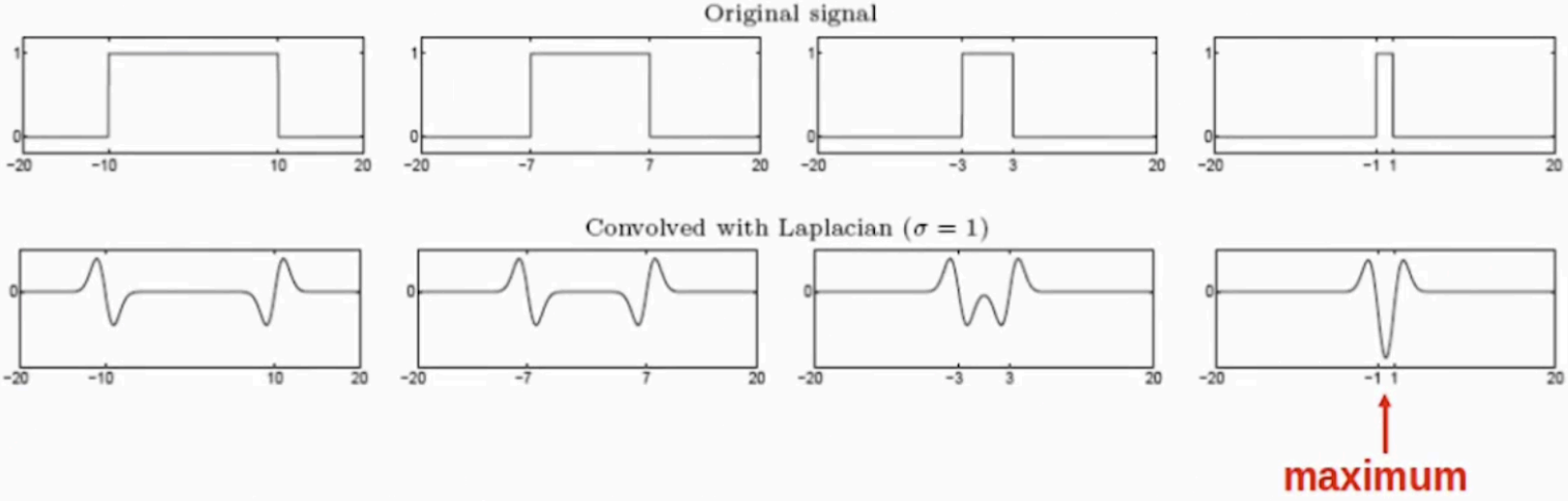


Courtesy: Forsyth and Ponce

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LoG on Blob

- Edge = ripple
- Blob = superposition of two ripples

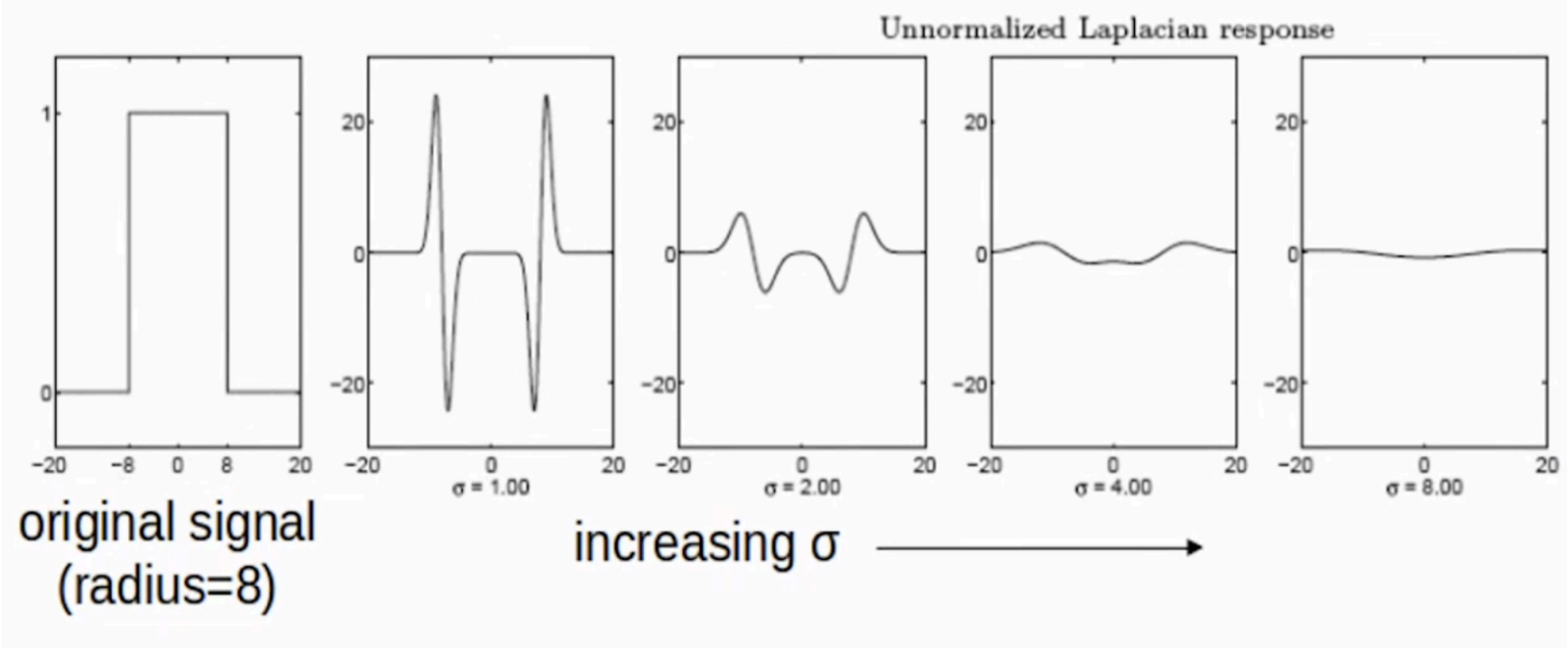


Spatial selection: The magnitude of the Laplacian response will achieve a maximum at the center of the blob, provided the scale of the Laplacian is matched to the scale of the blob.

Courtesy: Forsyth and Ponce

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Unnormalized LoG



Laplacian response decays as scale increases.

Courtesy: Forsyth and Ponce

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A blob is a double edge, examined in one dimension

A one-dimensional cross-section of a blob is a **pulse**, which rises and then falls, and so consists of two step edges placed back to back.

We already know that the LoG, the Mexican hat $\frac{d^2 G_\sigma}{dx^2}$, marks an edge by a **zero-crossing**.

Convolving the pulse with a LoG gives two responses.

- The rising edge produces a response of one sign.
- The falling edge produces a response of the opposite sign.

As the pulse **narrows**, or equivalently as the LoG **widens**, the two responses move toward each other and interact, until at one particular scale they merge into a single **extremum**.

That extremum, whether a maximum or a minimum, is the response of the blob. A bright blob on a dark background gives one sign and a dark blob on a bright background the other, and **both** are blobs.

The two roles are interchangeable. One may hold the blob fixed and vary the parameter σ of the LoG, which is what we do here, or hold σ fixed and vary the size of the blob. It is the adjustable σ of the LoG that provides the necessary control.

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Response of the LoG to a step edge

We take an ideal step $I(x) = u(x - x_0)$ and convolve it with a LoG, $\dfrac{d^2 G_\sigma}{dx^2}$, in which $G_\sigma(x) = \dfrac{1}{\sqrt{2\pi}\,\sigma}\, e^{-x^2/2\sigma^2}$.

We transfer one derivative onto the step, which is permissible because differentiation is linear and commutes with convolution:

$$u(x-x_0) * \frac{d^2 G_\sigma}{dx^2} = \frac{d}{dx}\Big[u(x-x_0)\Big] * \frac{dG_\sigma}{dx} = \delta(x-x_0) * \frac{dG_\sigma}{dx} = \frac{dG_\sigma(x-x_0)}{dx}$$

The derivative of the step is an impulse, and convolution with an impulse merely shifts the function. The response is therefore the first derivative of a Gaussian, centered at the edge.

Since $\dfrac{dG_\sigma}{dx} = -\dfrac{1}{\sqrt{2\pi}\,\sigma^3}\, x\, e^{-x^2/2\sigma^2}$, the response to the step edge is

$$R_\sigma(x) = -\frac{1}{\sqrt{2\pi}\,\sigma^3}\,(x - x_0)\, e^{-(x-x_0)^2/2\sigma^2} \qquad (1)$$

what happens at $x = x_0$?

The location and strength of the peaks

The peaks of (1) are found by setting $\frac{d}{dx} R_\sigma = 0$. Differentiating $-\frac{1}{\sqrt{2\pi}\sigma^3}(x-x_0)e^{-(x-x_0)^2/2\sigma^2}$ gives

$$-\frac{1}{\sqrt{2\pi}\,\sigma^3}\, e^{-(x-x_0)^2/2\sigma^2}\left[1-\frac{(x-x_0)^2}{\sigma^2}\right]=0 \quad\Longrightarrow\quad (x-x_0)^2=\sigma^2 \quad\Longrightarrow\quad x=x_0\pm\sigma$$

The two lobes lie **one σ on either side** of the edge, which is exactly the spacing shown in the figures. Substituting back into (1) gives

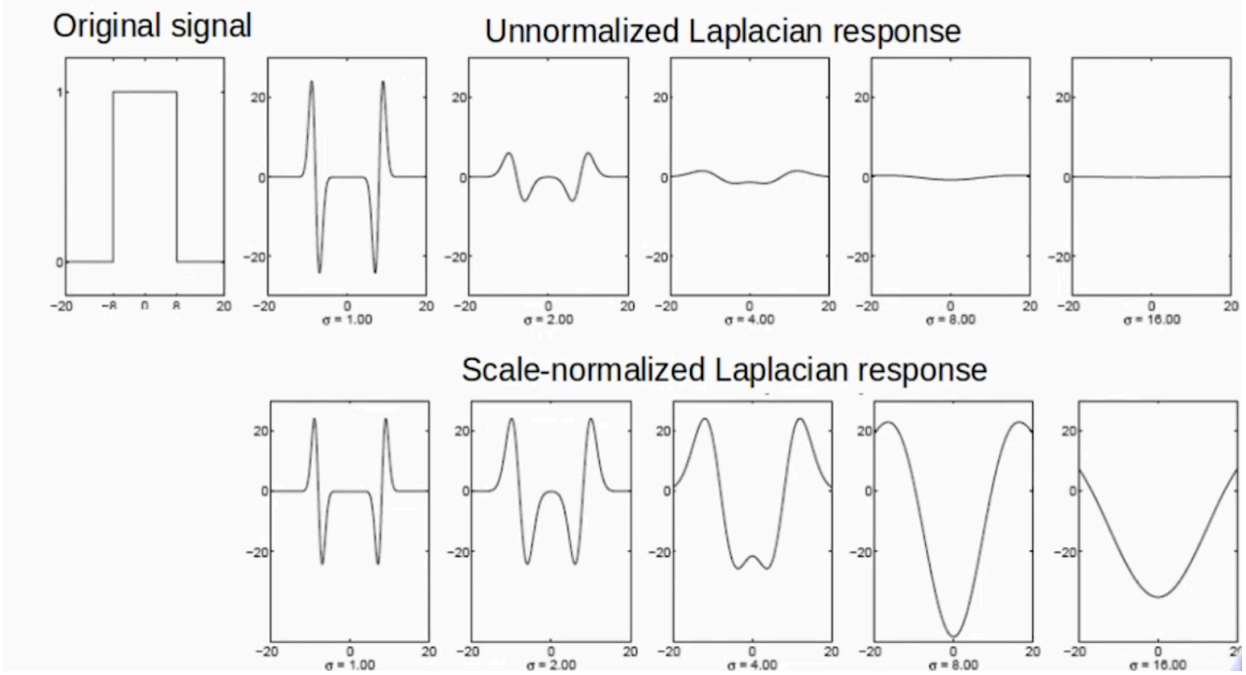
$$R_\sigma(x_0\pm\sigma) \;=\; \mp\frac{1}{\sqrt{2\pi}\,\sigma^2}\, e^{-1/2}$$

The peak **magnitude** is proportional to $\frac{1}{\sigma^2}$, and therefore **decays as the scale is coarsened**. An unnormalized LoG consequently cannot be compared across scales, since a larger σ is unfairly penalized.

The remedy is to multiply the LoG by σ^2. $\quad\boxed{\text{scale-normalized LoG} \;=\; \sigma^2\,\nabla^2 G_\sigma}$

The peak now takes the constant value $\mp\frac{1}{\sqrt{2\pi}}e^{-1/2}$, which is **independent of σ**, so that responses at different scales may at last be compared on equal terms.

The unnormalized and scale-normalized responses compared



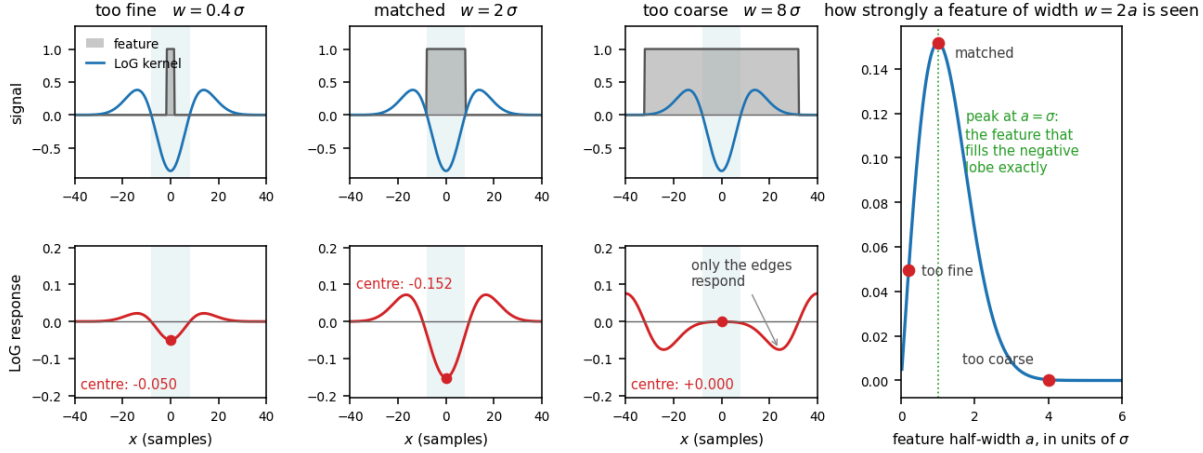
In the upper row the raw LoG response to a one-dimensional pulse decays as σ increases. In the lower row the factor σ^2 maintains the strength, so that a genuine **maximum** appears when the probe matches the pulse.

An interpretation of the decay in terms of area

Beyond the algebra there is a simple interpretation. A response is an **inner product** $\langle \text{signal}, \text{LoG} \rangle$, which is a signed area.

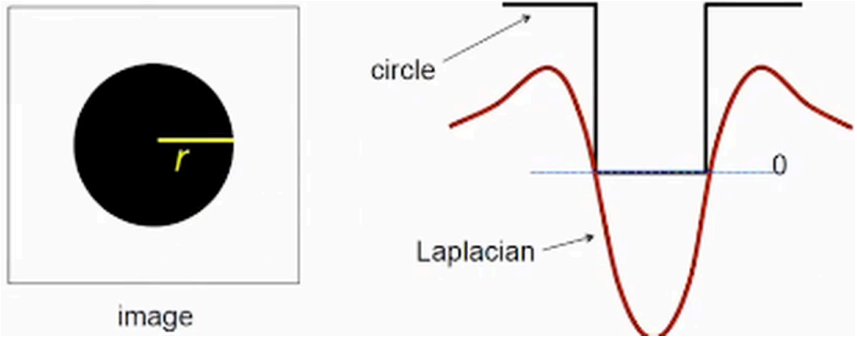
Consider a pulse slid against a fixed Mexican hat.

- Wherever the pulse is **zero**, the LoG contributes nothing, including its negative side-lobes, since it is multiplied by zero.
- Only the part of the pulse that overlaps the **positive central lobe** contributes to the sum.
- If the **probe is too large**, so that the blob is narrower than the central lobe, the zero part of the pulse extends beyond the lobe and the positive area **decreases**.
- If the **probe is too small**, so that the blob is wider, the **negative** side-lobes of the LoG fall on non-zero signal and subtract from the total.
- At the **best fit** the zero-crossings coincide with the edges of the blob, the area is entirely positive, and the response attains its **maximum**.



Scale selection: align the zeros of the Laplacian with the circle

At what scale does the Laplacian give a **maximum** response to a binary circle of radius r?



A cross-section through the center. At the best scale the two **zero-crossings** of the Laplacian sit exactly on the two edges of the circle.

The previous slide gave the condition: the response is largest when the **zero-crossings of the LoG land on the boundary of the blob**, so that the central lobe covers the blob and the side-lobes fall entirely on the background.

So we only need the **zeros** of the 2-D LoG.

$$\nabla^2 G(x,y) = \frac{x^2+y^2-2\sigma^2}{\sigma^4}\, e^{-\frac{x^2+y^2}{2\sigma^2}}$$

The exponential never vanishes, so the response is zero precisely on

$$x^2 + y^2 = 2\sigma^2,$$

a **circle of radius** $\sqrt{2}\,\sigma$ - the rim of the Mexican hat.

Setting that rim on the boundary of the blob, $\sqrt{2}\,\sigma = r$:

$$\sigma = \frac{r}{\sqrt{2}} \qquad \text{equivalently} \qquad \boxed{r = \sqrt{2}\,\sigma}$$

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Why the Gaussian is used: the scale-space axioms

Varying σ is meaningful only if coarsening the scale **removes** detail without **introducing** new detail.

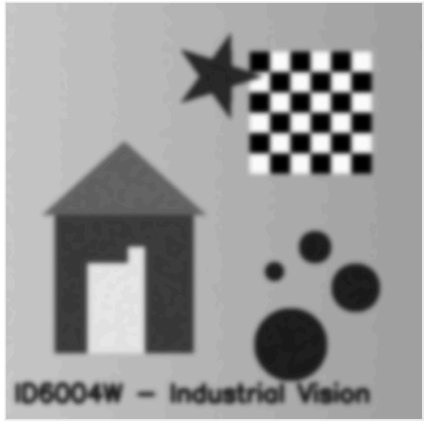
- **Causality**, or the requirement that no new structure be created. Moving to a coarser scale must not create extrema that were absent at finer scales, since otherwise detail would appear from nowhere.
- **Fine detail must dissolve monotonically** as the scale increases.
- Among all smoothing kernels the **Gaussian is the unique one** satisfying these conditions, a result established rigorously in one dimension by Lindeberg [12]. The LoG inherits the property directly, since the LoG is $\nabla^2 G$.

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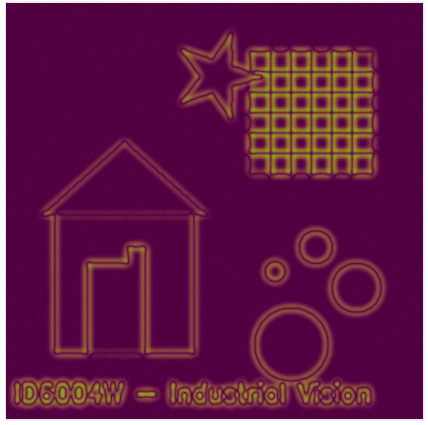
Demonstration of the scale-space LoG

image Synthetic scene *ideal step edges, exact corners — the textbook case*

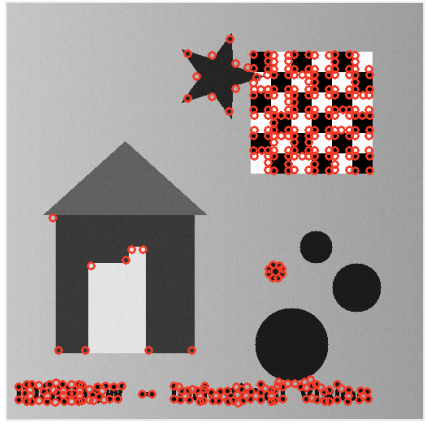
σ = 3.0 scale-normalize (× σ²) non-max suppression (r = √2 σ)



$G_\sigma * I$ — the smoothed image



|LoG response| — fixed brightness scale



detected blobs (circle radius = √2 σ)

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peak |resp| = 1.573
320 blobs
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Questions

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Summary

- The failure of the Harris detector, in that a magnified corner appears as a pair of edges, requires the introduction of a **scale axis**. Blobs provide a convenient feature on which to develop this.
- A blob is a **double edge**. The **LoG**, $\nabla^2 G_\sigma$, probes it and responds most strongly when its zero-crossings coincide with the boundary of the blob.
- The response to a step edge, equation (1), has peaks at $x_0 \pm \sigma$ with magnitude proportional to $1/\sigma^2$. Multiplying by σ^2 gives the **scale-normalized LoG**.
- A blob of radius r gives the strongest response at $\boxed{\sigma = r/\sqrt{2}}$, equivalently $r = \sqrt{2}\,\sigma$, which is what the circles in the output represent.
- **Scale space** is the image together with the axis σ. Keypoints are extrema of the magnitude in **all three dimensions**, and the **Gaussian** is the unique kernel that introduces no new structure across scale.

The next lecture treats SIFT. The LoG is expensive to compute, and Lowe approximates it by a **difference of Gaussians**. He constructs a Gaussian **pyramid** of octaves and attaches to each keypoint an orientation and a 128-dimensional **descriptor**, so that keypoints may be matched.

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References

1. Lindeberg, Tony. "Feature detection with automatic scale selection." International journal of Industrial Vision 30.2 (1998): 79-116.
2. Forsyth, David A., and Jean Ponce. Industrial Vision: a modern approach. prentice hall professional technical reference, 2002. 3.Marr, David, and Ellen Hildreth. "Theory of edge detection." Proceedings of the Royal Society of London. Series B, Biological Sciences (1980): 187-217.
3. Lindeberg, Tony. Scale-space theory in Industrial Vision. Vol. 256. Springer Science & Business Media, 2013.

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